Contact
jbrown1.618@gmail.com
(860) 271-9092

Joshua A. Brown

More
github.com/jbrown1618
linkedin.com/in/jbrown1618

Summary

Staff software engineer focused on AI products and developer platforms, with 12 years of software development experience. Hands-on full-stack technical leader in cross-team delivery, reusable architecture, and mentoring. Led cross-device Copilot experiences and frontend architecture serving 70 million daily views.

Technical Skills

Languages:	TypeScript, JavaScript, Go, Python
Front end:	React, HTML/CSS, WebAssembly, Accessibility, UI architecture, Jest, Integration testing
Backend & AI:	Node.js, API design, Cosmos DB, LLM integrations, Observability
Real time:	WebSockets, Azure Web PubSub, Cross-client synchronization
Leadership:	Cross-team delivery, Technical roadmaps, Architecture documentation, Mentoring, Stakeholder alignment

Professional Experience

GitHub/Microsoft *2020–Present*

Staff Software Engineer—GitHub Copilot

- Led architecture and phased delivery of Copilot Remote Control from zero to public preview in six weeks, coordinating platform, CLI, web, mobile, desktop, and IDE extension teams; preview reached 10K daily users and 30K remote sessions per day
- Defined forward- and backward-compatible session and event contracts for cross-client synchronization; led real-time communication enhancements using WebSockets via Azure Web PubSub
- Partnered with product and client teams on API requirements, milestones, and user-driven launch scope, balancing delivery velocity against long-term architectural rigor
- Delivered policy controls vital for enterprise adoption
- Implemented a dependency-graph linter enforcing hexagonal architecture and module boundaries as the service evolves from a single product into a multi-team platform
- Grew engineers' technical independence through modeling architectural decisions, collaborative design, and progressive ownership handoffs; mentees went on to lead major engineering initiatives
- Coordinated model-capacity planning across five teams. Built a unified usage dashboard and established forecasting standards to address rate-limiting risks

Senior Software Engineer—GitHub Repos and Copilot

- Architected the React rewrite of GitHub's file-viewing experience, serving 70 million daily views; enabled code-search integration with definition and usage navigation, substantially improved accessibility, and laid the foundation for later Copilot integration
- Designed explicit frontend extension points adopted by three product teams, enabling independent development in team-owned packages while preserving a cohesive product experience
- Architected an internal AI workflow platform combining LLM and deterministic steps in a directed acyclic graph; evaluations against comparable one-shot prompts demonstrated more consistent results
- Made Copilot Chat on GitHub.com model-agnostic, enabling its first multi-provider experiences; integrated repository, pull request, and issue context
- Documented architectural tradeoffs and rollout plans; maintained decision logs, including ADRs, to onboard seven engineers and align security and infrastructure partners

Software Engineer III—Azure DevOps

- Owned and optimized the Work Item Query Language and query auto-optimization engine, assuming responsibility for an area previously maintained by an entire team
- Improved Azure Boards accessibility and resolved live-site incidents supporting enterprise customer SLAs

SAS Institute—Data Visualization R&D *2017–2020*

Senior Associate Software Developer

- Built browser applications and libraries for SAS Viya using TypeScript, JavaScript, C++, and WebAssembly, including a customer-facing data visualization template editor
- Led development of an integration testing framework spanning JavaScript host code and a WebAssembly-based C++ runtime
- Drove adoption of shared testing and developer tools across teams through inner-source contributions and support

MetLife, Inc. *2014–2016*

Software Development Engineer II

- Developed an enterprise underwriting application supporting group benefits renewals for over 10,000 companies per month; rebuilt search to reduce end-to-end request time by a factor of 1,000
- Led engineers in building a Spring Boot and Angular proof of concept that became a standard for new projects; guided teams adopting the architecture
- Taught AngularJS and Play to senior engineers and automated more than a dozen recurring manual reports per month

Projects & Leadership

@josh-brown/vector

- Built and published a TypeScript linear algebra package on npm, implementing machine learning algorithms, matrix factorizations, and eigenvalue algorithms from scratch with zero runtime dependencies

Unicorn CLI

- Built a CLI for SAS's distributed build system and replaced manual steps with an automation-friendly workflow; engineers across the division adopted and contributed to the tool

Founder and Music Director, UNC Tarpeggios a cappella group

- Founded a university-recognized ensemble, recruited members, and led over 300 rehearsals and seven full-length concerts; managed production of a $10,000 studio album

Education

University of North Carolina at Chapel Hill

- B.S. in Applied Mathematics with Highest Distinction; B.A. in Physics; Minor in Music
- Chancellor's Carolina Scholarship: four-year full ride
- UNC-Chapel Hill National Merit Scholarship